

Roll No.

potential by the effect of temperature

TPE-301

B. TECH. (PE) (THIRD SEMESTER) END SEMESTER EXAMINATION, Jan., 2023

THERMODYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are saturated pressure and temperature ? Draw the P-V-T behavior of pure fluids and discuss the terms saturated phase and superheated phase from the diagram. (CO1, CO2)
- (b) Define the following : (CO1, CO2)
 - (i) Point and path function with examples
 - (ii) Intensive and Extensive Properties with examples
 - (iii) Degree of freedom
 - (iv) Enthalpy and Internal energy
 - (v) Second law of thermodynamics
 - (vi) Entropy of the system

P. T. O.

- (c) What are the two propositions involved in the Carnot principle ?
Discuss Ideal gas as the Carnot engine working substance. (CO3)
2. (a) Illustrate Adiabatic process and hence derive an expression for the work done under adiabatically condition of the system. Specify the symbol used. (CO3, CO4)
- (b) Define polytropic process. What is the process called when value of $n = 1$ and $n = \infty$? Hence derive the equation for work done, internal energy and heat supplied. (CO1, CO4)
- (c) What do you mean by COP of a refrigerator ? Establish relationship between COP of refrigerator and Efficiency of a heat engine. (CO1, CO5)

Solve :

It is required to freeze 1 kg water at 273 K by means of a refrigeration machine which operates in the surroundings at 300 K. The latent heat of fusion of ice at 273 K is 334.11 kJ/kg. Determine :

- (i) The minimum amount of work required
- (ii) The heat given up to the surroundings
3. (a) Classify the thermodynamics properties of pure fluids. How are Maxwell's equations helpful ? Derive the Maxwell's equations from the exact differential equations of the four energy properties. (CO1)

- (b) Establish the entropy-heat capacity relationships and hence determine the increase in entropy of solid magnesium when the temperature is increased from 300 K to 800 K at atmospheric pressure. The heat capacity is given by the equation as :

(CO1, CO2)

$$C_p = 26.04 + 5.586 \times 10^{-3}T + 28.476 \times 10^{-4}T^{-2}$$

- (c) What is the importance of Clapeyron equation ? Explain the Clausius-Clapeyron equation.

(CO2, CO5)

Solve : Calculate the vapor pressure of water at 363 K, if the vapor pressure at 373 K is 101.3 kPa. The mean heat of vaporization in this temperature range is 2275 kJ/kg.

4. (a) Explain Gibbs free energy and hence derive the modified equations for internal energy and enthalpy by using the energy and Maxwell's relations. Specify the symbol used.

(CO1, CO5)

- (b) Define the inversion point of Joule Thompson co-efficient (μ). Prove that the Joule Thompson co-efficient (μ) in Throttling Process can be expressed by the expression as :

(CO4, CO5)

$$\mu = \frac{\left[T \left(\frac{dV}{dT} \right)_P - V \right]}{C_p} \text{ constant } h$$

- (c) Explain fugacity and fugacity coefficient. Derive the expression for the determination of fugacity of pure gases by using compressibility factor Z.

(CO1, CO6)

P. T. O.

5. (a) Explain chemical potential. Deduce an expression on the chemical potential by the effect of temperature. (CO4, CO6)

(b) Discuss "The Linde process" for gas-liquefaction along with the component construction arrangements (diagram) of equipments used for the process. (CO4, CO5)

(c) What do you mean by refrigerant ? To design the refrigeration process the type of refrigerant is very important. What are the characteristics of parameter on which the choice of refrigerant for the refrigeration process is to be considered ? Discuss in detail. (CO2, CO6)

Roll No.

TPE-302

B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
APPLIED GEOLOGY
Time : Three Hours
Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is physical weathering ? What are the factors affecting physical weathering ? (CO1)
- (b) What is exogenic and endogenic processes ? List and describe the *three* exogenic processes. (CO1)
- (c) List and describe the *four* landforms developed due to river erosion with the help of a diagram. (CO1)
2. (a) What is chemical weathering ? What are the process through which weathering takes places ? (CO2)

P. T. O.

(2)

- (b) What are the 7 crystal system ? Describe. (CO2)
- (c) What is meant by mineral ? What is difference between amorphous solid and crystalline solid ? Describe the physical properties of mineral use in hand specimen identification of mineral. (CO2)
3. (a) What is igneous rock ? Discuss the process of formation of igneous rock. (CO3)
- (b) What is rock cycle ? Explain. (CO3)
- (c) What is Nesosilicate, Sorosilicate, Cyclosilicate and Inosilicate ? Explain and give an example of mineral of each group. (CO2)
4. (a) Draw the geological time scale with major boundaries. (CO4)
- (b) What are the laws of stratigraphy ? Discuss. (CO4)
- (c) Describe the Litho-stratigraphic and Bio-stratigraphic classification. (CO4)
5. (a) What is stress and strain ? Discuss the importance of folding in petroleum industry. How can you recognize fold ? (CO5)
- (b) What is difference between fractures and fault ? Discuss the geometric classification of fault. (CO6)
- (c) Describe the classification of fold on the basis of appearance in cross-section, symmetry fold, interlimb angle and attitude of the fold. (CO6)

Roll No.

TPE-303

B. TECH. (PE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

FLUID AND PARTICLE MECHANICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Show that the equation $s = ut + \frac{1}{2}at^2$ is dimensionally homogeneous.

(CO1)

(b) Derive Bernoulli equation by integrating Euler equation of motion.

(CO1)

(c) Explain how can you measure vacuum pressure with the help of a U-tube manometer.

(CO1)

P. T. O.

(2)

2. (a) What is head ? How can Bernoulli equation be modified for viscous flow ? (CO1, 2)
- (b) Show that the velocity profile in a steady laminar flow in a circular pipe is parabolic. (CO1, 2)
- (c) What are major and minor losses ? How can they be estimated ? (CO1, 2)
3. (a) What are Pi terms in Buckingham Pi theorem ? How can they be constructed ? (CO2, 3)
- (b) Explain how dimensional analysis can be beneficial for comparing the performance of scale model with the prototype. (CO2, 3)
- (c) What are Froude's number and Mach's number ? Give *one* example of their use each. (CO2, 3)
4. (a) Explain the different parts of a centrifugal pump with diagram and their working. (CO3, 4)
- (b) Derive the expression for rate of water/fluid delivered by a reciprocation pump. (CO3, 4)
- (c) Draw a diagram and explain the various parts of a double acting reciprocating pump. (CO3, 4)
5. (a) Explain fluidization with diagram and indicate its use. (CO5, 6)
- (b) Derive the shear stress distribution for laminar flow in a circular pipe. (CO5, 6)
- (c) What is screening ? Explain effectiveness of a screen. (CO5, 6)

Roll No.

TPE-304

B. TECH. (PE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

GEOMATIC ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the various classifications of survey. (CO1)
- (b) Differentiate between plane and geodetic surveying. (CO1)
- (c) Define surveying. Discuss in brief the principles of surveying. (CO1)
2. (a) Describe with the help of sketches the characteristics of contour. (CO2)

P. T. O.

- (b) The following are the fore and back bearings of the sides of a closed traverse : (CO2)

Side	FB	BB
AB	150° 15'	330° 15'
BC	20° 30'	200° 30'
CD	295° 45'	115° 45'
DE	218° 00'	38° 00'
EA	120° 30'	300° 30'

Calculate the interior angles of the traverse.

- (c) What are the different errors in theodolite work ? How are they eliminated ? (CO2)
3. (a) Discuss the advantages and disadvantages of plane table surveying over other methods. (CO3)
- (b) Determine the length of BC and its bearing : (CO3)

Line	Length, m	Bearing
AB	180	80° 30'
BC	?	?
CD	160	280° 30'
DE	170	195° 30'
EA	200	15°

- (c) Explain different methods of plane table surveying. (CO3)

(3)

4. (a) A section line AB appears to be 10.16 cm on a photograph for which the focal length is 16 cm. The corresponding line measures 2.54 cm on a map which is to a scale 1/50,000. The terrain has an average elevation of 200 m above mean sea level. Calculate the flying altitude of the aircraft above mean sea level when the photograph was taken. (CO4)
- (b) Define Photogrammetry. Explain the advantages of Photogrammetry. Discuss the use and application of Photogrammetry. (CO4)
- (c) Explain the types of aerial photographs with neat sketch. (CO4)
5. (a) Explain the Electromagnetic Radiation (EMR) interactions with the atmosphere and the earth surface. (CO5)
- (b) Write a short note on Geographic Information system (GIS). (CO5)
- (c) Explain Remote sensing process in detail. (CO5)

Roll No.

TMA-307

**B. TECH. (PE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

ADVANCED MATHEMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Express the function $f(x)$ as a Fourier integral representation and

hence evaluate $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$, where $f(x) = \begin{cases} 1, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$

(CO2)

(b) Find the Fourier sine integral for the function $f(x) = e^{-kx}; x \geq 0$, where k is any positive constant. Hence show that

$$\int_0^\infty \frac{\sin \lambda x}{k^2 + \lambda^2} d\lambda = \frac{\pi}{2} e^{-kx}. \quad (\text{CO2})$$

P. T. O.

- (c) Find the Fourier transform of : (CO2)

$$f(x) = \begin{cases} a^2 - x^2, & |x| \leq a \\ 0, & |x| > a \end{cases}$$

2. (a) Show that the function $f(z) = z|z|$ is not analytic anywhere. (CO1)
- (b) Evaluate the complex integral $\int_{1-i}^{2+3i} (z + z^2) dz$ along the line joining the points $(1, -1)$ and $(2, 3)$. (CO1)
- (c) Using Cauchy's integral formula, evaluate $\int_C \frac{2z^2 + z}{z^2 - 1}$, where C is the circle $|z - 1| = 1$. (CO1)
3. (a) Find the bilinear transformation which maps the points $z = -i, 0, i$ in the z -plane onto the points $w = -1, i, 1$ in the w -plane. (CO3)
- (b) Show that $e^x(x \cos y - y \sin y)$ is a harmonic function. Find the analytic function for which $e^x(x \cos y - y \sin y)$ is an imaginary part. (CO3)
- (c) Define the singularities and poles with the help of an example. (CO3)
4. (a) If the probability density function of a continuous random variable X is given by : (CO6)

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance and standard deviation.

- (b) The number of airplane landings at an airport in a minute (X) and their probabilities are given by :

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X . (CO6)

- (c) The distribution of the number of road accidents per day in a city is Poisson with mean 4. Find the number of days out of 100 days when there will be (i) No accident, (ii) At least 2 accidents. (CO6)
5. (a) Define the Fuzzy set and membership function with the help of an example. (CO5)
- (b) Following data depicts the statistical values of rainfall and production of wheat in a region for a specified time period : (CO5)

	Mean	Standard Deviation
Production of Wheat (kg/unit area)	10	8
Rainfall (cm)	8	2

Estimate the production of wheat when rainfall is 9 cm if correlation coefficient between production and rainfall is given to be 0.5.

(4)

TMA-307

(c) Define the following :

(CO5)

- (i) Complement of a fuzzy set
- (ii) Product of two fuzzy sets
- (iii) Power of a fuzzy set
- (iv) Normal fuzzy set

TMA-307

50

Roll No.

TPE-311

B. TECH. (PETROLEUM ENGINEERING)

(THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

INTRODUCTION TO PETROLEUM OPERATIONS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the different geophysical methods of exploration ? (CO1)
(b) Describe the classification of wells used in oil industry. (CO1)
(c) What is meant by well planning ? What are the flow path of well planning ? (CO1)
2. (a) What is oil based mud ? What are the advantages and disadvantages of using oil based mud ? (CO2)

P. T. O.

(2)

- (b) What is meant by directional drilling and deviation control ? What are the applications of directional drilling ? (CO3)
- (c) What are the component of block and tackle system in a rotary drilling ? Discuss. (CO3)
3. (a) What is well logging ? Mention its major applications. Describe subsurface condition around a borehole. (CO4)
- (b) What is a drill time log ? What are its use ? What are the factors influencing drill time log ? (CO4)
- (c) What is open and cased hole logging ? With the help of a diagram, describe calliper log tool and its working principle. (CO4)
4. (a) What is difference between kick and blow out ? What is hydrostatic pressure and formation pressure ? What are the components of well control operation ? (CO4)
- (b) What are the modes of transportation of crude oil ? Discuss. (CO6)
- (c) How can the crude oil be distilled ? What are process involved during the distillation of crude oil ? (CO6)
5. (a) What is meant by perforation operations in an oil well ? What are the different types of perforation ? (CO5)
- (b) What are the advantages and disadvantages of cased add perforated completion over open hole completion ? (CO5)
- (c) What are the products obtained during the distillation of crude oil ? (CO6)